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ORIGINAL ARTICLE

Detecting Local Myocardial Spatiotemporal Repolarization Gradients With Clinical Mapping Arrays

Tasnia Subha, BSc; Stéphane Massé, MASc; Yusuf Abderrahman, MEng; Golnaz Mokhtar-Sasani; Patrick F.H. Lai, MSc; John Asta; Christopher Labos, MD; Abhishek Bhaskaran, MD, PhD; Praloy Chakraborty, MD; Vijay S. Chauhan, MD; Paul Dorian, MD; Kumaraswamy Nanthakumar, MD

BACKGROUND: Activation recovery interval (ARI), extracted from unipolar electrograms, serves as a practical surrogate for repolarization during experimental studies in vivo. Far-field signal contamination and low spatial resolution obscure regional repolarization gradients and duration alternans detection using unipolar ARI. We hypothesized that the attenuation of far-field contamination with the principal component-referenced unipole will allow for a more accurate assessment of true local repolarization gradients and spatially assess action potential duration alternans.

METHODS: Unipolar ARI and the novel method, Repol^{Loc}, were validated for the detection of spatial and temporal repolarization changes using simultaneous optical and electrical mapping in a rabbit Langendorff model. Repolarization changes were created using global infusion of ibutilide or pinacidil, or topical application of lidocaine. Epicardial mapping was conducted in a porcine Langendorff model following the topical application of lidocaine to investigate the spatial resolution of each method. Generalized linear models of the two methods were used to compare with optical action potential duration (APD80).

RESULTS: Following the infusion of antiarrhythmic drugs, the Repol^{Loc} method (slope=0.90) had a slightly higher correlation to optical APD80 than the ARI method (slope=0.79). Following regional application of lidocaine, Repol^{Loc} was better able to localize the site of drug administration with an average 26.12% reduction as compared with 18.66% reduction in unipolar ARI (P=0.0046). Additionally, temporal repolarization alternans and restitution changes assessed by Repol^{Loc} method tracked optical APD80 quantified time domain changes.

CONCLUSIONS: Repol^{Loc} has higher sensitivity to local spatiotemporal repolarization heterogeneities and alternans than traditional ARI. Although ARI only correlates with uniformly distributed changes in repolarization in the entire myocardium, Repol^{Loc} also provided accurate regional gradient assessment and duration alternans of repolarization. These findings suggest ARI has significant far-field contamination and Repol^{Loc} may provide a better clinical mapping tool for spatiotemporal repolarization gradient mapping.

GRAPHIC ABSTRACT: A graphic abstract is available for this article.

Key Words: electrophysiologic study ■ epicardial mapping ■ intracardiac electrogram ■ lidocaine ■ pinacidil

Current approaches for ventricular arrhythmia ablation focus on depolarization mapping. Late activation and other substrate-based methods neglect reentrant arrhythmias that are driven and maintained by repolarization gradients and heterogeneity.[1] The importance of quantification of repolarization is highlighted in the mechanism for ventricular tachycardia (VT) or ventricular fibrillation (VF), which requires sufficient heterogeneity

Correspondence to: Kumaraswamy Nanthakumar, MD, FRCPc, Division of Cardiology, University Health Network, Toronto General Hospital, 150 Gerrard St W, GW3-526, Toronto, ON, Canada, M5G 2C4, Email kumar.nanthakumar@uhn.ca; or Tasnia Subha, BSc, University Health Network, Toronto General Hospital, 150 Gerrard St W, GW3-526, Toronto ON, Canada, M5G 2C4, Email tasnia.subha@uhn.ca
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WHAT IS KNOWN?

- Repolarization gradients and temporal repolarization dispersion are key arrhythmic substrates and can trigger reentry.
- Unipolar activation recovery interval is insufficient in assessing local spatial heterogeneities in repolarization.
- Unipolar activation recovery interval is not practically useful in the cath lab for spatiotemporal duration alternans assessment.

WHAT THE STUDY ADDS

- The novel Repol^{Loc} method provides a true local assessment of repolarization in the X-Y plane of the myocardium and changes in time domain.
- The Repol^{Loc} method can be used to identify areas of steep spatiotemporal repolarization gradient, which are vulnerable to reentry initiation and could be relevant for localizing critical sites for reentry using contemporary clinical mapping arrays.

Nonstandard Abbreviations and Acronyms

APD	action potential duration
APD80	80% of estimated optical action potential duration
ARI	activation recovery interval
AT	activation time
dV/dt	voltage-time derivative
Repol^{Loc}	local repolarization assessment method
RT	repolarization time

and gradient in local repolarization in space and in time for reentry to occur.[2] Additionally, R-from-T and R-to-T phenomenon are critically dependent on steep gradients of repolarization for reentry to initiate and sustain.[3] These mechanistic bases of arrhythmias indicate the need for an accurate mapping tool to detect local heterogeneities in repolarization and local gradients in repolarization to modulate for clinical benefit. There is currently no known tool available for such usage.

Additionally, the major challenge to testing the hypothesis regarding delivery of treatment to regions of steep gradients is the inability to clinically identify local changes in repolarization with clinical mapping catheters. Preclinically, optical mapping is a gold-standard method for measuring local action potential duration and can provide spatial assessment of repolarization unlike monophasic action potential or premature stimuli.[4] However, optical mapping cannot be used clinically due to dye toxicity concerns and the use of excitation-contraction uncouplers to prevent signal distortion.[4] Therefore, the current candidates for repolarization assessment in a clinical setting are limited by the shortcomings of the unipolar and bipolar electrograms. The current methods for identifying time of repolarization in unipolar electrograms are the Wyatt activation recovery interval (ARI) method and an alternative method which identifies the repolarization time (RT) based on the voltage-time derivative (dV/dt) of T-waves depending on polarity. However, these methods do not deal with the morphology of the T wave that resembles far-field components of the unipolar signal. As such, both of these methods have been shown to not correlate well to gold-standard methods for measuring action potential duration ex vivo.[5,6] Over and above inability to detect local T waves, the major limitation for both of these methods is the spatial integration of recorded unipolar signals, confounding true local repolarization assessment. Though far-field integration is thought to be less on a bipolar electrogram, we recently demonstrated the fallacy of this dogma.[7] In addition to this, given the directional bias, the T-wave is often not visible in bipolar electrograms as tissue anisotropy plays a significant role in the direction of T-wave propagation.[8]

In addition to spatial heterogeneities of repolarization, temporal dispersion has also been shown to be a relevant marker of arrhythmogenicity.[9–11] Action potential duration (APD) alternans has been associated with T-wave alternans and thus, arrhythmogenesis.[9–11] Furthermore, spatially discordant alternans has been shown to be especially relevant to vulnerability to reentry.[9,12] Current methodology for measuring APD alternans include monophasic action potentials[13–15] which are limited spatially, and optical mapping[12,16] which is not clinically feasible. Orini et al[17] elegantly demonstrated in the operating room on cardiopulmonary bypass the use of unipolar ARI to globally map APD alternans. However, this has not been translated to mapping catheters. As such, there remains a need for a mapping methodology that can provide local spatiotemporal assessment of repolarization heterogeneity in the cath lab.

The advent of equi-spaced arrays with 3-dimensional electrode arrangements have made alternative electrogram constructs possible that can address some of the shortcomings of unipolar and bipolar electrograms. We have recently demonstrated that principal component-referenced unipoles improve detectability of local myocardial activation by attenuating far-field contamination and are independent of direction.[7] We tested the hypothesis that principal-component referenced unipoles will allow for accurate spatiotemporal detection of true local repolarization as a result of the attenuation of far-field contribution by validating it against transmembrane potential.

METHODS

The data that support the findings of this study are available from the corresponding author on request. All experiments were approved by the Animal Care Committee at the University Health Network, Toronto and adhere to the ARRIVE guidelines.

Principal Component-Referenced Local Repolarization (Repol^{Loc})

The purpose of the local repolarization assessment method (Repol^{Loc}) signal is to eliminate the far-field contribution to a traditional unipolar electrogram. For this analysis, the unipolar electrograms were collected from 2-dimensional grid electrode catheters (OPTRELL mapping catheter, Biosense Webster, Irvine, CA). To derive the principal component-referenced signals for each electrode, the algorithm takes a set of adjacent 9 neighbors (organized in a 3×3 on the electrode grid) and creates a signals matrix *A* (Figure 1A). Principal component analysis is then applied to the matrix *A*. Principal component analysis is a method that decomposes the signals into multiple components and is often used for denoising. The first principal component estimates the directions in the data where the differences are most noticeable. We use the reconstructed signals from the first component extracted from principal component analysis to obtain a reconstructed matrix, called matrix B, of the same dimension as matrix A. The derived signals from matrix B are then averaged to derive a principal component reference; Figure 1A). The principal component reference contains mostly far field activity that is common to all 9 electrograms. Finally, the principal component reference is subtracted from the unipolar electrogram from the middle electrode in the 3×3 group to create the near field electrogram, Repol^{Loc} (Figure 1A). This

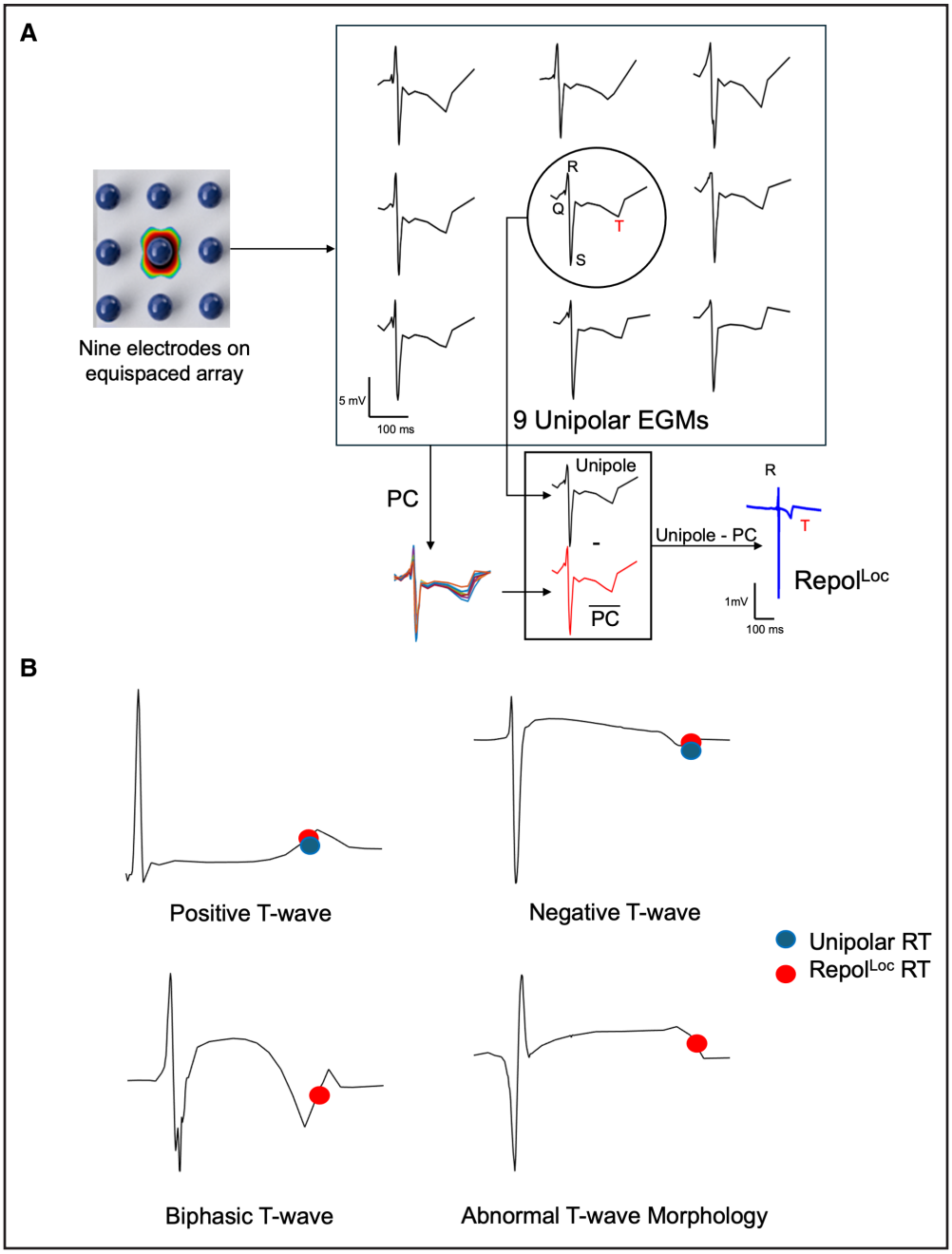


Figure 1. Signal processing methodologies for repolarization time (RT) measurement.
A, Derivation of the Repol^{Loc} signal for the central unipolar electrogram (EGM) using principal component (PC) analysis. **B**, Unipolar RT vs Repol^{Loc} method on various T-wave morphologies.

process eliminates the common element between the neighboring electrodes and only the local information remains to form the Repol^{Loc} signal.

Activation Time Method

The standard method for annotating activation time (AT) in unipolar electrograms is the maximum negative dV/dt on the QRS.[18] This method was used for both unipolar and Repol^{Loc} signals. The mean AT values were used for both methods to assess the validity of AT as a time reference for APD estimation in each model.

RT Methods

The traditional method for measuring RT from unipolar EGMs is the Wyatt ARI method, which takes the maximum positive dV/dt on the unipolar T-wave as the time of complete repolarization regardless of T-wave polarity.[18] We proposed a novel method to annotate RT on the Repol^{Loc} signal, which takes the maximum absolute dV/dt on the T-wave regardless of T-wave morphology (Figure 1B). This is to address the greater variation between Repol^{Loc} signals in T-wave polarity, as well as abnormal morphologies that are difficult to annotate. These methods were used to estimate APD from the respective associated electrograms which can be compared with true local APD values.

Simultaneous Optical and Electrical Mapping on Ex Vivo Rabbit Model

Healthy rabbit hearts were explanted from 12 New Zealand white rabbits (≈3.5 kg) and arranged into a Langendorff model perfused in Tyrode solution, potentiometric dye (0.25 µmol/L Rh237), and excitation-contraction decoupler (3.4 µmol/L Blebbistatin; Figure 2A). To alter repolarization uniformly

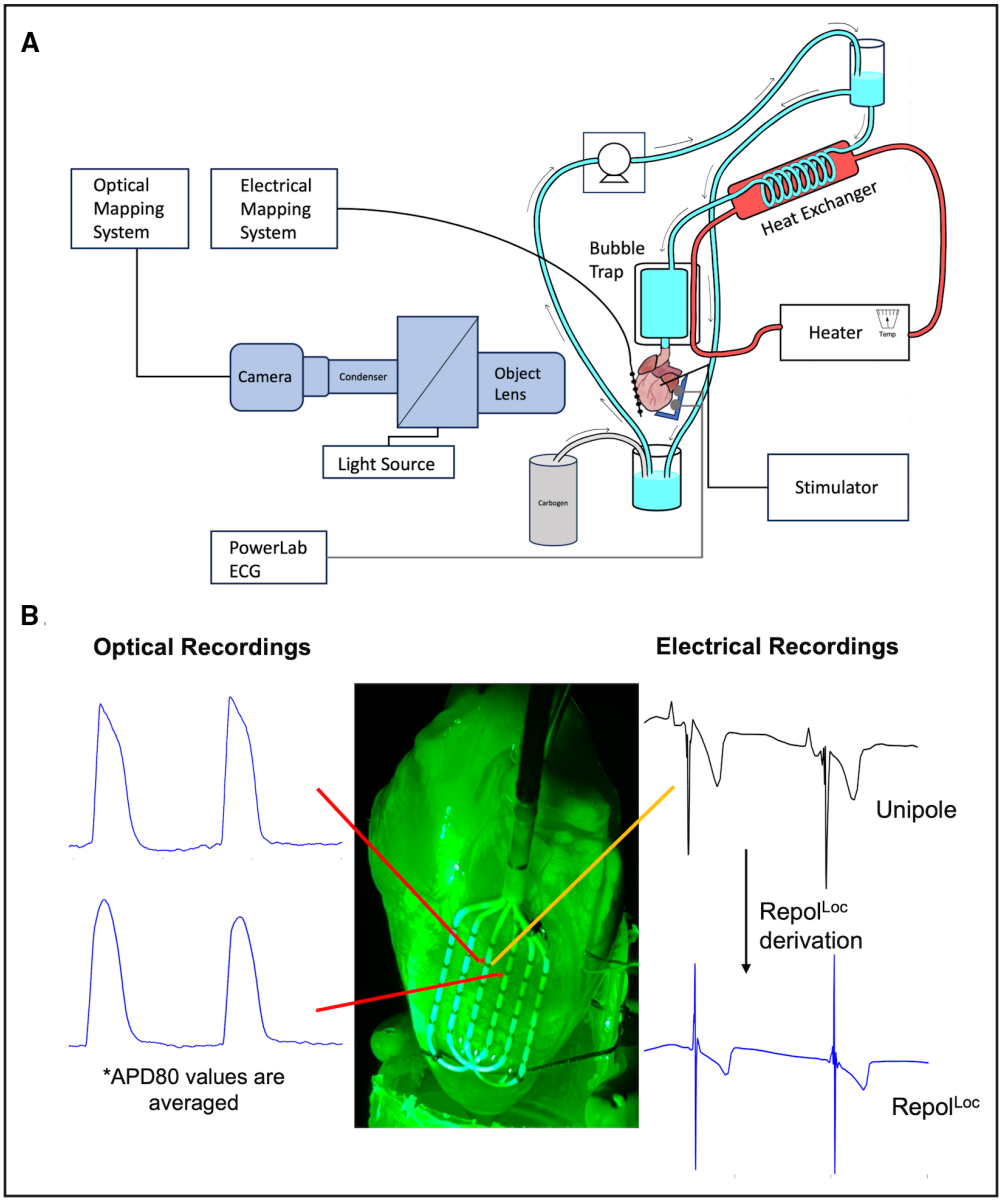


Figure 2. Experimental set-up for simultaneous optical and electrical mapping.
A, Diagram of the rabbit Langendorff and optical mapping set-up. **B**, Comparison of fluorescence and electrical signals.

throughout the myocardium, 0.5 and 1 µmol/L of ibutilide or 20 and 30 µmol/L of pinacidil were added to the perfusate and circulated in the Langendorff for 5 to 15 minutes before mapping. To alter repolarization regionally, 0.1 mL of 4% lidocaine was applied to the myocardium near the bottom of the electrode array using a Hamilton syringe. The temperature of the perfusate was regulated at 37 °C. The 6×6 OPTRELL mapping catheter (Biosense Webster, Irvine, CA) was placed to the anterior epicardial surface covering both ventricles. Contact between the electrode array and the myocardial surface was ensured noninvasively using suture wire to tie the array against the heart. For fluorescence mapping, the dye fluorescence was excited using a xenon light source through a 530 nm green filter. The emission light beam was filtered through a pass with 715 nm length. Fluorescence was recorded using the Ultima-L high-speed CMOS camera (SciMedia Ltd, Costa Mesa, CA). The hearts were electrically stimulated through a pair of silver electrodes placed epicardially near the base of the right ventricle and from the right ventricle from the endocardial apex. Voltage fluorescence signals were recorded simultaneously to epicardial electrical signals from the grid catheter with incremental pacing, capturing signals both during pacing and at the termination of pacing at each frequency (Figure 2B). We also performed an extrastimulus pacing protocol on 2 rabbit hearts for assessing Effective Refractory Period. 80% of estimated optical action potential durations (APD80) from the voltage fluorescence signals were analyzed with a custom software written in Matlab (The Mathworks Inc., Natick, MA) and compared with unipolar ARI and the derived Repol^{Loc} values from the electrode array.

Epicardial Mapping on Ex Vivo Swine Model

Healthy swine hearts were explanted from 5 Yorkshire pigs (50–60 kg) and arranged in a Langendorff set-up as described above. The 6×8 OPTRELL mapping catheter (Biosense Webster, Irvine, CA) was placed to the anterior epicardial surface of the left ventricle. Contact between the electrode array and the myocardial surface was ensured using a gauze wrap to secure the grid. To alter local repolarization, 0.2 mL and 0.4 mL of 2% lidocaine was applied to the myocardium at the center of the electrode array using a Hamilton syringe. ARI was measured for the unipolar EGMs obtained from the electrode array using the Wyatt method. The Repol^{Loc} signals were derived from the unipoles and the corresponding APD estimates were annotated using the Repol^{Loc} method.

Statistical Methods

Paired comparisons between optical APD80 and electrical measurements were presented in a correlation plot. To compare the association between optical APD80 and the two methods, generalized linear regressions were performed and compared with the identity line. To account for repeated measurements within each experiment, a hierarchical mixed-effects linear regression model was also used to compare the two electrical methods and optical APD80. The hierarchial 3-level model was used with the different pacing frequencies nested within the different conditions (baseline versus drug intervention) nested within each experiment. To identify bias in the two methods compared with the gold-standard, Bland-Altman analysis was performed using the differences between the methods and the gold standard against the average of the values. To tune APD values to the electrical methods, Bland-Altman analyses were performed using the differences between the methods and various optical APD percentages. Percentages were used to present discrete changes where a gold-standard method was not feasible. Wilcoxon test was used to compare discrete changes before and after drug intervention. Mann-Whitney *U* test was used to compare percent changes near versus far from drug administration site following drug intervention. Categorical analysis of T-wave detectability was performed with a Fisher exact test. Statistical analysis was performed using GraphPad Prism 10 (GraphPad Software, Boston, MA).

RESULTS

Validating Repol^{Loc} With Optical Mapping and Comparison to Unipolar ARI

Perfusion of ibutilide resulted in APD prolongation, shown in the increase in optical APD80 on the epicardial surface. Perfusion with pinacidil induced significant APD shortening, shown in the decrease in optical APD80 on the epicardial surface. Incremental epicardial pacing before and after drug administration created further variations in APD. The linear relationship between the Repol^{Loc} method and optical APD80 ($Y=0.9030\times X+7.358$, $R^2=0.8411$) was closer to the identity line than the relationship between unipolar ARI and optical APD80 ($Y=0.7888\times X+20.62$, $R^2=0.7439$; Figure 3A). After adjusting for repeated measurements with the hierarchical models, the Repol^{Loc} method more closely matched optical APD80 (mean difference, −7.18 ms [95% CI, −8.13 to −6.22]) than did the ARI method (mean difference,−11.43 ms [95% CI, −12.40 to −10.46]; Table). The difference between Repol^{Loc} and unipolar ARI was small but statistically significant (4.25 ms [95% CI, 5.23–3.27]; $P<0.001$; Table). Overall, the repolarization changes that were apparent in the optical signals were more accurately detected using Repol^{Loc} method than the ARI method.

Detectability of Local Changes in Repolarization With ARI Compared With Repol^{Loc}

In the rabbit model, the application of lidocaine created a localized decrease in optical APD in the bottom right quadrant of the electrode array (Figure 4A). The Repol^{Loc} map better localizes this effect at the location of lidocaine application as validated by the optical APD80 map than the ARI map (Figure 4A).

In the pig Langendorff model, the application of lidocaine shortened the APD at the site of administration, at the center of the electrode array. This was reflected in the significant reduction in ARI using the Repol^{Loc} method at the site of administration (mean 26.12% reduction; $P<0.0001$; Figure 4B). This reduction in APD was not as apparent from the change in ARI at the

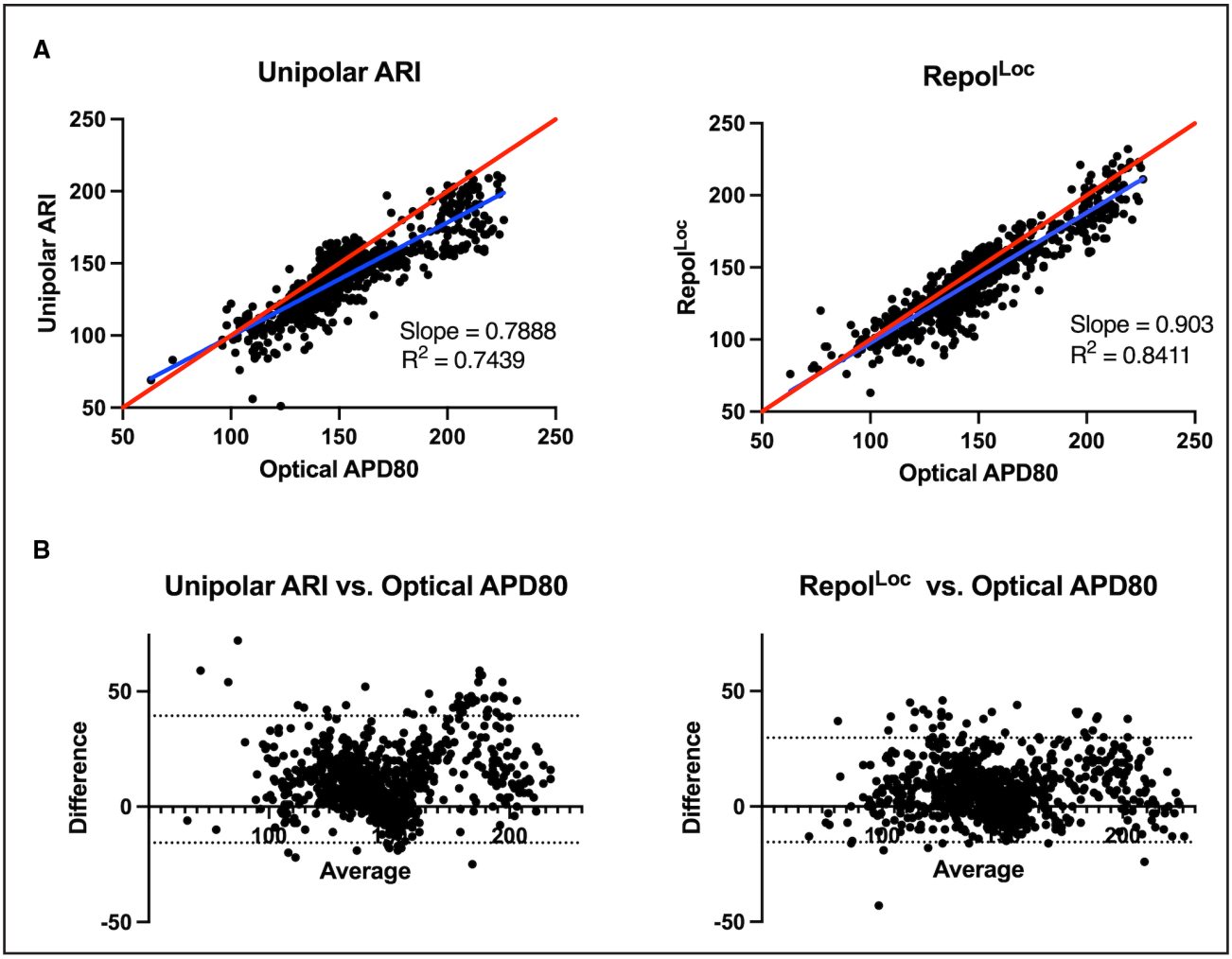


Figure 3. Validation of unipolar ARI and Repol^{Loc} against optical APD80 values.
Unipolar ARI, Repol^{Loc}, and APD80 values measured from baseline, post-pinacidil, and post-ibutilide recordings following epicardial pacing in the rabbit Langendorff model (n=11). **A**, Correlation plots and generalized linear regressions comparing unipolar and Repol^{Loc} values to corresponding optical APD80. **B**, Bland-Altman plots comparing unipolar and Repol^{Loc} values to corresponding optical APD80.

site of lidocaine administration (mean, 18.66% reduction; $P<0.0001$; Figure 4B). The difference in percent reduction between the two methods at the site of drug administration was statistically significant ($P=0.0046$). A greater difference in percent change near versus far from the application site was observed in the Repol^{Loc} signal (11.94%; $P<0.0001$) compared with the unipolar EGM (6.77%; $P=0.0022$).

Tuning APD to Repol^{Loc}

To explore the possibility that the Repol^{Loc} signal is better representative of other percentages of the optical APD, we performed a Bland-Altman comparison with various APDxx values where xx = 30, 50, 80, and 90. APD80 most closely aligns with both the unipolar ARI (11.89 bias [95% CI, −15.62 to 39.41]) and the Repol^{Loc}

Table. Comparison between unipolar ARI, Repol^{Loc}, and optical APD80 using a hierarchical mixed-effects linear regression model

Comparison	Mean difference(ms)	SE*	Significance	95% CI†
Repol^{Loc} vs optical APD80	−7.1756	0.48708	$P<0.001$	−8.1303 to −6.2210
ARI vs optical APD80	−11.4277	0.49546	$P<0.001$	−12.3988 to −10.4566
Repol^{Loc} vs ARI	−4.2520	0.50138	$P<0.001$	−5.2347 to −3.2693

APD80 indicates 80% of estimated optical action potential duration; and ARI, activation recovery interval.
*Standard error.
†Confidence intervals.

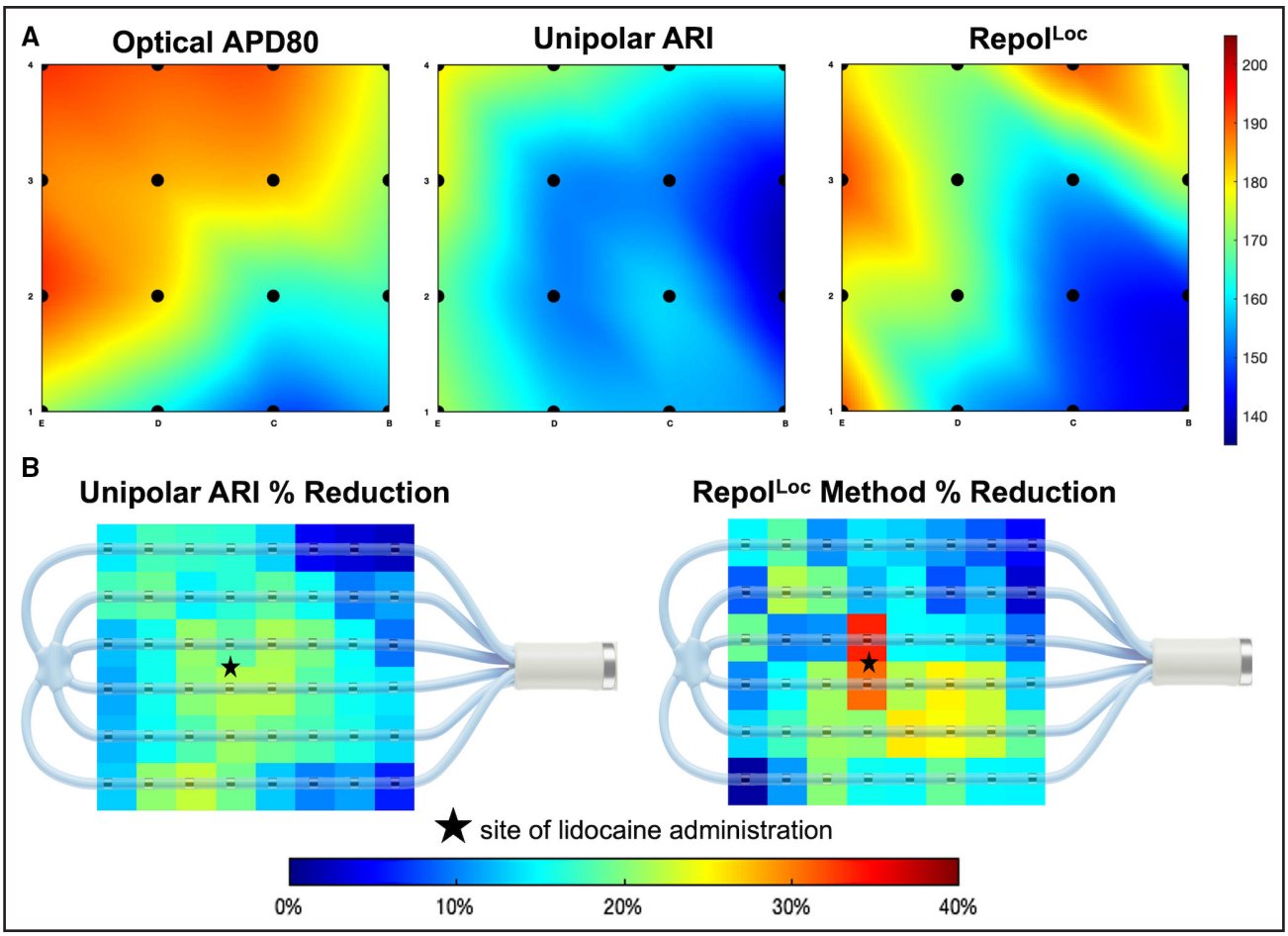


Figure 4. Local repolarization shortening using lidocaine.
A, Optical APD80 map and corresponding unipolar ARI and Repol^{Loc} maps following topical application of lidocaine to rabbit Langendorff. **B**, Percent reduction in ARI vs Repol^{Loc} after application of lidocaine from epicardial mapping on pig Langendorff.

method (7.26 ms bias [95% CI, −15.4 to 29.91]; Figure 5). Therefore, APD80 is an accurate and stable estimation of optical APD given that APD90 is heavily influenced by baseline wander and noise (Figure 5).

Validating Detection of Local Temporal Repolarization Dispersion

Following global infusion of ibutilide, we occasionally observed APD alternans in the optical signal during 6.0 Hz pacing (Figure 6). This suggests that this occurrence of alternans is occurring locally, in the superficial layer of the myocardium. The alternans was also apparent in the Repol^{Loc} signal but not in the unipolar signal which suggests that the far-field attenuation in the Repol^{Loc} allows for local spatial detection of APD alternans (Figure 6). Additionally, we also compared rate response of repolarization (restitution curve) measurements using extrastimulus using the Repol^{Loc} approach compared with optical mapping. The detection of restitution changes in repolarization using the Repol^{Loc} method following varied extrastimulus intervals is comparable to the changes observed in optical APD80 (Figure 7).

Practical Detectability With Repol^{Loc}

To identify the specific conditions where Repol^{Loc} is most effective for repolarization assessment, we compared T-wave detectability between different pacing locations and controls versus drug intervention. All unipolar recordings had large, detectable T-waves. However, in ~12.60% of Repol^{Loc} signals the T-wave was not detectable despite the presence of a QRS. Changes in pacing location has an effect on the detectability of T-waves. Following epicardial pacing, the T-wave was visible in 95.83% of Repol^{Loc} signals. However, following endocardial pacing, the T-wave was only visible in 79.17% of Repol^{Loc} signals (Figure 8A). This discrepancy suggests that there was a clear association between pacing location and T-wave detectability (P=0.0007) and T-wave detection in Repol^{Loc} may be limited to epicardial wavefronts and subsequent repolarization gradients

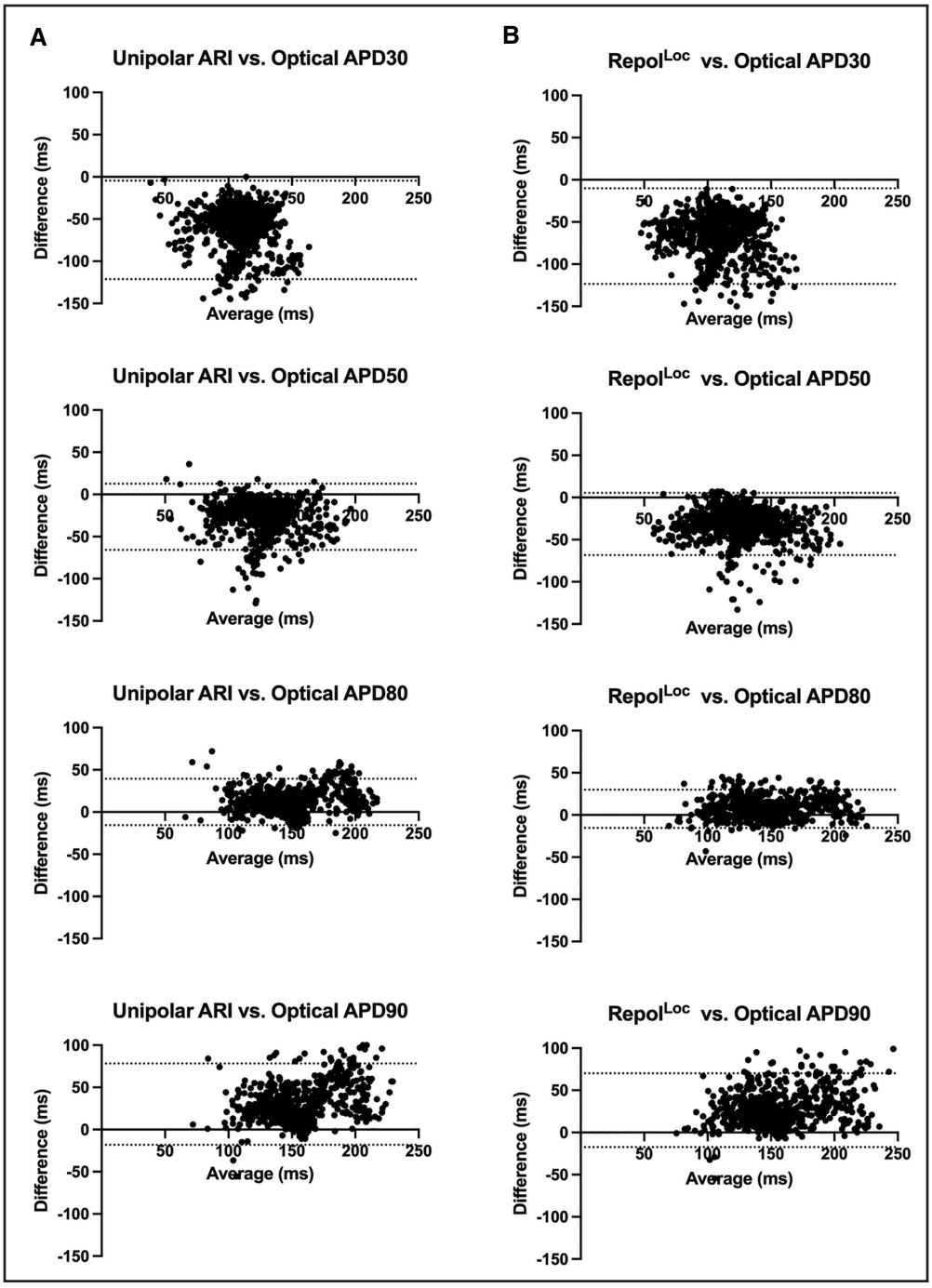


Figure 5. Bland Altman comparisons of the electrical methods against APDxx values where xx=30, 50, 80, and 90. **A**, APDxx vs unipolar ARIs. **B**, APDxx vs Repol^{Loc}. Optical APDxx indicates xx% of estimated optical action potential duration; and ARI, activation recovery interval.

within the X-Y plane. A comparison of T-wave detectability before and after the administration of antiarrhythmic drugs shows that pre-drug intervention T-waves were detectable in 92.02% of the signals whereas after drug intervention T-waves were detectable in 83.33% of signals (Figure 8B). Therefore, the administration of drugs was also associated with T-wave detectability (P=0.0185).

Contribution of AT to APD Estimation Differences

The differences in AT between the unipolar and Repol^{Loc} signals were negligible following epicardial pacing for both the rabbit and porcine models. In the rabbit model, the mean difference in AT was 2.94 ms (SEM=2.105) between the two methods. In the porcine model, the

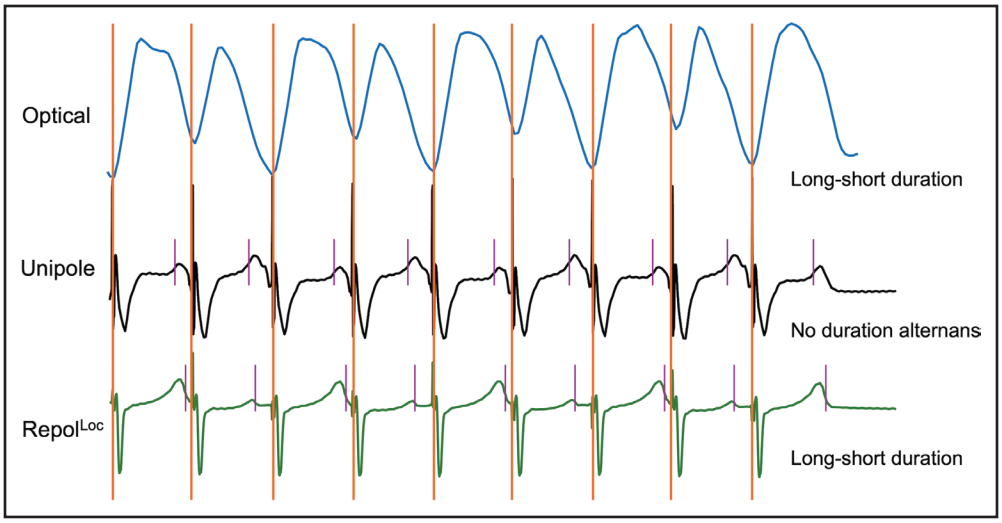


Figure 6. Comparison of local action potential duration (APD) alternans between an optical, unipolar, and RepolLoc signal.

mean difference in AT was 1.673 ms (SEM=0.2637) between the two methods. Therefore, the estimated APD values from these two methods can be used as an assessment of repolarization heterogeneity.

DISCUSSION

In this study, we validated the novel RepolLoc method against optical APD80 to detect local spatial and temporal changes in repolarization within the surface of the myocardium using a clinical mapping array catheter. The RepolLoc method detected both spatial repolarization gradient and APD alternans, which was not apparent from traditional unipolar ARI, validated by detection using optical APD80.

The RepolLoc Method for RT Measurements

Repolarization changes validated using changes in optical APD80 values are more accurately detected using the novel RepolLoc methodology than the traditional ARI. There is a higher degree of correlation between true APD80 values from optical mapping and RepolLoc than with ARI. Given that RepolLoc signal processing has been shown to alter QRS morphology,[7] changes in T-wave morphology from the traditional unipolar T-wave were expected. To address the effect of signal processing on the polarity and overall morphology of the T-wave, the novel RepolLoc method takes the maximum absolute dV/dt value regardless of T-wave polarity. This methodology builds on the current technique for differentiating near-field and far-field signals: evaluating the steepness or dV/dt.[19] The RT measurements from RepolLoc signals are assumed to be more local based on previous findings that show the attenuation of far-field contribution when the first principal component is subtracted.[7,20] Several experimental and clinical studies have demonstrated the role of spatial repolarization dispersion on the induction and maintenance of polymorphic VTs, such as Torsades de Pointes arrhythmias.[21–23] These regions of repolarization heterogeneity are not detectable via methods that integrate far-field information such as ECG imaging.[24] In comparing the current intracardiac methodology for repolarization assessment, unipolar ARI, to optical APD values, we further demonstrated the limitations of the unipolar electrogram in detecting local repolarization which have been previously investigated.[5,6] Thus, the need for the novel RepolLoc method is the improved spatial accuracy of RT detection, as validated by optical APD, which is crucial for detection of the repolarization heterogeneity substrate associated with reentrant VTs and arrhythmogenic disorders such as long-QT syndrome.[21–24]

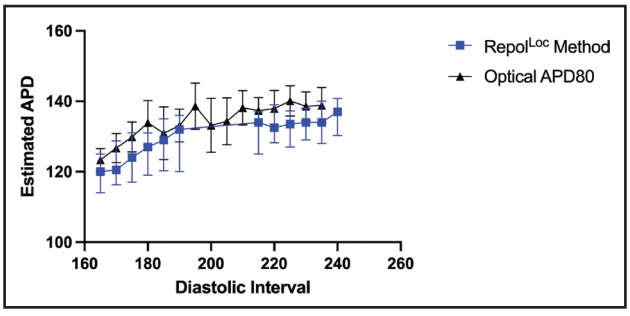


Figure 7. Restitution curves for optical APD80 and the RepolLoc method following various extrastimulus intervals.

Importance of Detection of Local Repolarization Gradient

The added value of the RepolLoc method is the ability to specifically detect spatial and temporal differences in repolarization, well-known markers of arrhythmogenicity.[3,21–23,25] Repolarization heterogeneity is associated with a mechanism of reentry initiation called the R-on-T phenomenon.[26] This mechanism of reentry initiation

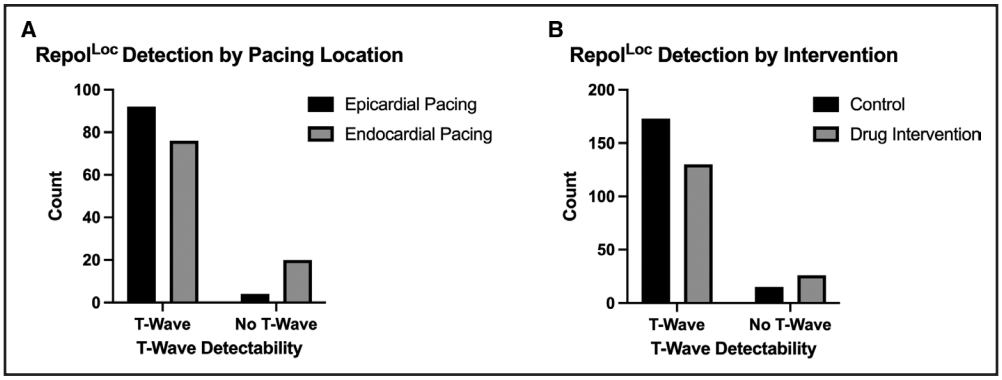


Figure 8. Comparison of T-wave detectability in the rabbit model.
A, Repolarization time detectability during epicardial vs endocardial pacing . **B**, Repolarization time detectability control vs drug intervention. (n=11).

can occur either when a premature ventricular complex arises focally and propogates to an area of delayed repolarization or arises directly from the T-wave as a result of a repolarization gradient.[26] The premature ventricular complex can initiate a tachycardia if there is significant repolarization heterogeneity forming unidirectional block, facilitating functional reentry.[1,2] Particularly, a spatial repolarization pattern where regions of cardiomyocytes with short RTs are connected to regions with long RTs as a result of abnormalities in potassium voltage-gated channel expression is associated with block and thus the formation of a reentrant circuit.[1] Inducibility of polymorphic VTs is associated with these spatial RT gradients and the effect they have on local restitution curves.[21,27,28] This has lead to the incorporation of RT measurement and repolarization mapping into the evaluation of reentry vulnerability to identify VT sites of origin.[2,29] However, thus far, this methodology has not significantly improved the identification of critical VT sites compared with other arrhythmia substrates and is limited by far-field contamination in the unipolar repolarization mapping aspect.[2,29] This limitation of unipolar repolarization mapping is further demonstrated by the finding that ARI does not accurately spatially delimit areas of prolonged or shortened repolarization based on the comparison with our optical APD data. Attenuation of far-field contribution in Repol^{Loc} allows for more accurate and localized detection of repolarization heterogeneity which will significantly improve spatial resolution for repolarization mapping.

Nuances of Repol^{Loc} for Repolarization Assessment

In many cases, T-wave detectability is diminished in Repol^{Loc} even when the corresponding unipole has a large T-wave. Repol^{Loc} T-wave detectability is diminished following endocardial pacing and slightly less detectable after a drug (lidocaine, pinacidil, ibutilide) is administered.

Endocardial pacing implies a stronger Z-axis component to the wavefront than during epicardial pacing. During epicardial pacing, similar to the activation sequence, repolarization proceeds from epicardium to endocardium as a result of the epicardial APD being intrinsically shorter than endocardial APD, producing a transmural gradient.[30,31] This spatial and temporal alignment of activation and recovery fronts results in a steeper, more coherent repolarization wavefront, which is reflected in a sharper and more uniform T wave morphology. In contrast, endocardial pacing produces opposing activation and repolarization sequences due to the intrinsic transmural gradient.[30,31] These effects likely reduce the explanatory power of the principal component during endocardial pacing and introduce variability in local repolarization timing. Consequently, the synchronized activation-repolarization pattern during epicardial pacing enhances the signal-to-noise ratio of the T wave and improves the accuracy and robustness of the Repol^{Loc} T wave. Therefore, the decreased detectability of the T-wave following endocardial pacing suggests that the Repol^{Loc} signal may not be as useful for transmural wavefronts when activation and repolarization sequences are opposed. This suggests a need for an electrogram that can differentiate signal components within the intramural space, particularly when evaluating the recovery of transmural wavefronts.

Presence of Repol^{Loc} T-wave indicates there are differences in the RT and T-wave morphology in each group of 9 neighboring electrodes that do not get eliminated when subtracting the average first principal component from the 9 signals. This itself suggests the existence of spatial repolarization heterogeneity. However, since other factors such as signal gain, wave direction, and signal-to-noise ratio also affect T-wave detectability, it cannot be reliably used as an indicator of spatial repolarization gradient.

Insights and Nuances of Repolarization Alternans Detection

The attenuation of far-field in the Repol^{Loc} method allows for the detection of local APD alternans validated by the

optical signal. The unipolar signal is mostly oblivious to the alternans due to far-field integration from regions with no alternans. This suggests greater accuracy in the spatiotemporal detection of APD heterogeneities, which has previously been shown to be a critical marker of arrhythmogenicity.[9,12] APD alternans mapping has previously been investigated intraoperatively using unipolar electrograms from an epicardial sock.[17] The novel Repol^{Loc} method allows for such mapping while improving spatiotemporal accuracy in a clinical mapping array during electrophysiology cath lab mapping procedures. Additionally, the Repol^{Loc} method may be used to detect restitution dynamics with comparable accuracy to optical mapping, which has been implicated in the generation of discordant alternans.[32]

The limitation of the Repol^{Loc} method on the transmural axis is further corroborated by an observation we made of an example discordance in APD alternans between the optical signal and the Repol^{Loc} signal (Figure S1). The unipole being oblivious to either pattern of alternans suggests integration of both patterns (Figure S1). Given that the work by Denham et al suggests the principal component-referenced signals integrate some far-field information from the transmural axis, the alternans detected by the Repol^{Loc} signal being discordant to optical APD alternans from the superficial layer of the myocardium suggests that the discordance might be within the transmural axis (Figure S1).[7] Since the Repol^{Loc} method cannot detect the depth from which the alternans originates, an alternative method is needed to localize the origin of temporal repolarization dispersion within the intramural space.

Study Limitations

A limitation with the porcine Langendorff model used to evaluate the local repolarization assessment of the two methods is the lack of a gold-standard reference such as optical mapping due to issues with size and tissue properties of the pig heart. As such, the interpretation of this data is reliant on accurate documentation of the site of drug application, assuming there was no run off or spread following topical application.

To address this limitation, we conducted simultaneous mapping in a rabbit model, which allowed for validation of the methods with optical APD as a reference. However, given the size of the model, it is difficult to define to accurately define near versus far-field and replicable the production of spatial repolarization gradient, hence, the focus on the effect of drugs administered via global infusion in this model. The field of view for optical mapping was also limited to the location of the epicardial surface where we could place the electrode array with maximal contact.

Additionally, AT values from each method were not compared with the time of activation in the optical signals. While the differences in AT values between the methods were small relative to estimated APD values during epicardial pacing, it is possible that different wavefront directions could amplify differences in AT between the unipolar and principal component-referenced signal.

Clinical Need

Current success rates of both reentrant ventricular arrhythmia ablation remain modest, with significant recurrence rates. Additionally, substantial knowledge gaps exist, particularly in rapidly evolving channelopathies, idiopathic VT, and VF. The current clinical paradigm for substrate mapping relies heavily on measuring depolarization and slow conduction, rather than repolarization. As current techniques do not assess spatiotemporal repolarization heterogeneity, clinicians lack the means to measure spatial differences in repolarization and time domain changes within the mapped myocardium. Therefore, there remains a pressing need for better mapping approaches that allow for effective diagnosis, risk prediction, and successful ablation therapies based on repolarization gradients and APD alternans, critical determinants of reentry. Detecting spatiotemporal repolarization gradients accurately is critical for risk stratification of VT/VF substrates and for designing targeted therapies, including catheter-based or even gene/cell-based approaches.

Potential Clinical Innovation

Given that repolarization gradients and APD heterogeneity are key substrates and at times trigger of reentry and therefore arrhythmogenicity, a clinical mapping system that spatially maps repolarization and APD would be a valuable innovation. We propose here a method coupled to a clinical mapping array that could be significant for identifying arrhythmogenic substrate. To target repolarization abnormalities for ablation therapy, we propose the innovative use of equi-spaced electrode arrays for near real-time assessment of repolarization gradients in the electrophysiology cath lab. Local detection of repolarization gradients could be useful for risk stratification of VT substrates and identifying key substrates in long QT and Brugada syndromes. In addition, it would serve to monitor proarrhythmic potential of anti-arrhythmic therapies. Using 2-dimensional electrode arrays to populate a Repol^{Loc} map will provide local estimations of APD over the X-Y plane of the myocardium allowing for identification of steep APD gradients in real-time. An alternate method in development for assessing local transmural repolarization gradient that builds on the orthogonal bipole approach previously validated by Massé et al may be used in conjunction with Repol^{Loc} to assess gradient in all axes of the myocardium.[33–35] These maps would be useful to stratify VT substrates, as a diagnostic tool for long QT and Brugada syndrome, and possibly as a

tool for identification of target sites for ablation therapy. Mapping efforts in previous clinical studies have been retrospective, manually annotated, and limited to data from decapolar linear catheters.[36,37] Using commercially available equi-spaced electrode arrays, we have conducted spatiotemporal analysis with novel computational methods which allows for accurate local repolarization measurements that can be plotted in unit space on the myocardial surface similar to isochronal maps. We hope this tool will allow clinicians to prospectively evaluate repolarization gradient mapping to localize critical substrates that may improve clinical outcomes of ablation therapies.

Conclusions

The Repol^{Loc} method is more accurate for detecting local spatiotemporal changes in repolarization than the traditional ARI method and is validated by true optical APD80 values. This method allows for the accurate detection of spatial repolarization dispersion and duration alternans in the X-Y plane of the myocardium, which are key predictors of reentry initiation and arrhythmogenicity. This method can be deployed in the clinical mapping cases that use mapping arrays that are designed with close equi-spaced electrodes.

ARTICLE INFORMATION

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Affiliations

University of Toronto (T.S., A.B., P.C., V.S.C., P.D., K.N.). The Hull Family Cardiac Fibrillation Management Laboratory, University Health Network, Toronto, ON (T.S., S.M., Y.A., G.M.-S., P.F.H.L., J.A., K.N.). McGill University, Montreal, QC, Canada (C.L.).

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Dr Nanthakumar is a consultant for Biosense Webster. The other authors report no conflicts.

Supplemental Material

Figure S1

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